

$$g(x) = \frac{\sin x}{1 + \tan x}$$

$$u = \sin x \quad u' = \cos x$$

$$v = 1 + \tan x \quad v' = \frac{1}{\cos^2 x}$$

$$g(x) = \frac{\cos x (1 + \tan x) - \frac{1}{\cos^2 x} (\sin x)}{(1 + \tan x)^2}$$

Simplifying:

$$\text{Let } L = \cos x (1 + \tan x) - \frac{1}{\cos^2 x} (\sin x)$$

$$L = \cos x + \cos x \tan x - \frac{\sin x}{\cos^2 x}$$

$$\Rightarrow \tan = \frac{\sin x}{\cos x}$$

$$L = \cos x + \frac{\cos x \cdot \sin x}{\cos x} - \frac{\sin x}{\cos^2 x}$$

$$L = \cos x + \frac{\sin x}{\cos x} - \frac{\sin x}{\cos^2 x}$$

$$L = \cos x + \sin(x) \left(\frac{1}{\cos x} - \frac{1}{\cos^2 x} \right)$$

$$g'(x) = \frac{\cos x + \sin x \left(\frac{1}{\cos x} - \frac{1}{\cos^2 x} \right)}{(1 + \tan)^2}$$

